

THERMAL POWER STATION

A **THERMAL POWER STATION** IS A POWER STATION IN WHICH HEAT ENERGY IS CONVERTED TO ELECTRICITY.

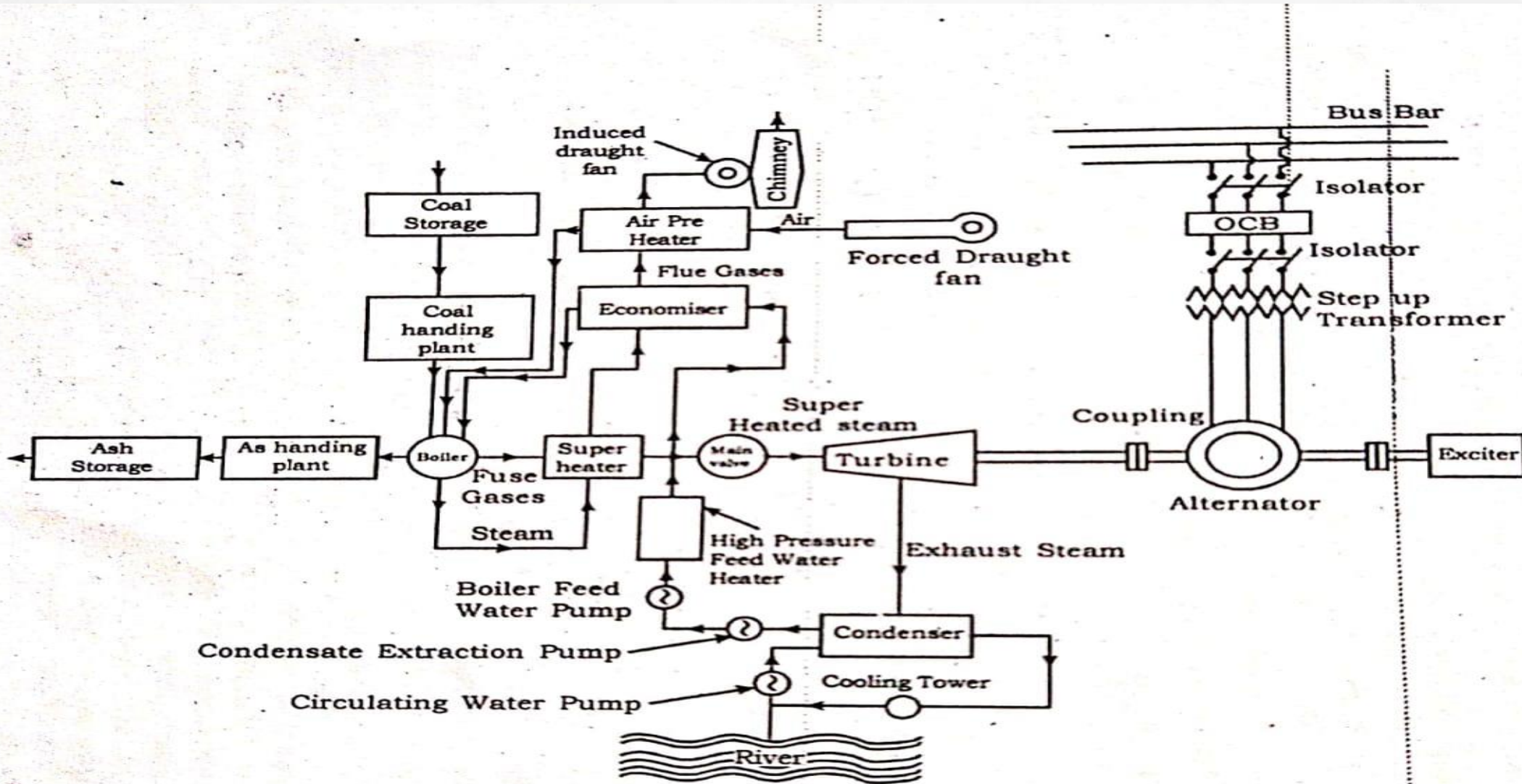
- TYPICALLY, FUEL IS USED TO BOIL WATER IN A LARGE PRESSURE VESSEL TO PRODUCE HIGH-PRESSURE STEAM, WHICH DRIVES A STEAM TURBINE CONNECTED TO AN ELECTRICAL GENERATOR.
- THE LOW-PRESSURE EXHAUST FROM THE TURBINE PASSES THROUGH A STEAM CONDENSER AND IS RECYCLED TO WHERE IT WAS HEATED. THIS IS KNOWN AS A RANKINE CYCLE. NATURAL GAS CAN ALSO BE BURNT DIRECTLY IN A GAS TURBINE SIMILARLY CONNECTED TO A GENERATOR.

- WATER POWER STATIONS (WHICH GENERATE HYDROELECTRICITY) ARE EXCLUDED FROM THIS CATEGORY, SINCE THEY CONVERT THE POTENTIAL ENERGY OF WATER INTO ELECTRICITY VIA A WATER TURBINE.
- THE DESIGN OF THERMAL POWER STATIONS DEPENDS ON THE INTENDED ENERGY SOURCE: FOSSIL FUEL, NUCLEAR POWER, BIOFUEL.
- CERTAIN THERMAL POWER STATIONS ARE ALSO DESIGNED TO PRODUCE HEAT FOR INDUSTRIAL PURPOSES; OR DESALINATION OF WATER, IN ADDITION TO GENERATING ELECTRICAL POWER.

GLIMPSE OF THERMAL POWER STATION



WORKING



❖ **ADVANTAGES:**

- LESS INITIAL COST AS COMPARED TO OTHER GENERATING STATIONS.
- IT REQUIRES LESS LAND AS COMPARED TO HYDRO POWER PLANT.
- THE FUEL (I.E. COAL) IS CHEAPER.
- THE COST OF GENERATION IS LESSER THAN THAT OF DIESEL POWER PLANTS.

❖ **DISADVANTAGES:**

- IT POLLUTES THE ATMOSPHERE DUE TO THE PRODUCTION OF LARGE AMOUNT OF SMOKE. THIS IS ONE OF THE CAUSES OF GLOBAL WARMING.
- THE OVERALL EFFICIENCY OF A THERMAL POWER STATION IS LOW (LESS THAN 30%).